

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

Treemap Layout Functions and Constants

Visualization Data Preparation Functions

File Data Preparation Functions

File Data Preparation

Data Preview

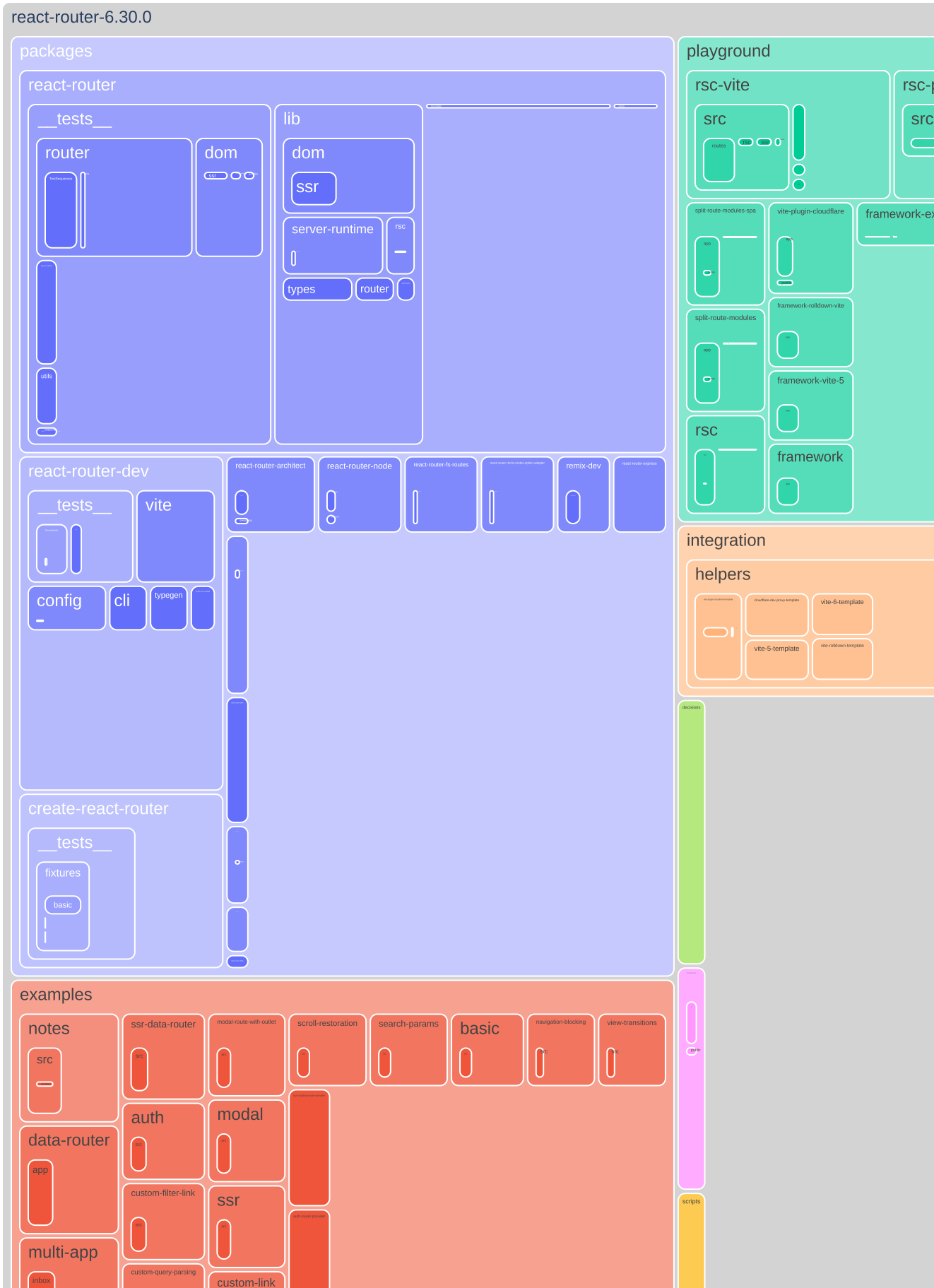
	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	232.000000	232.000000	232.000000	232.000000	232.000000	232.000000
mean	20.383621	12.086207	605.818966	107.594828	358.146552	165.556034
std	94.815950	36.880072	3199.780891	214.166823	459.171478	292.409332
min	1.000000	2.000000	2.000000	0.000000	0.000000	0.000000
25%	2.000000	3.000000	18.000000	2.000000	50.750000	6.500000
50%	5.000000	6.000000	61.500000	39.000000	148.500000	38.000000
75%	12.000000	10.000000	225.250000	103.000000	364.000000	149.000000
max	1304.000000	491.000000	40088.000000	1263.000000	1782.000000	1316.000000

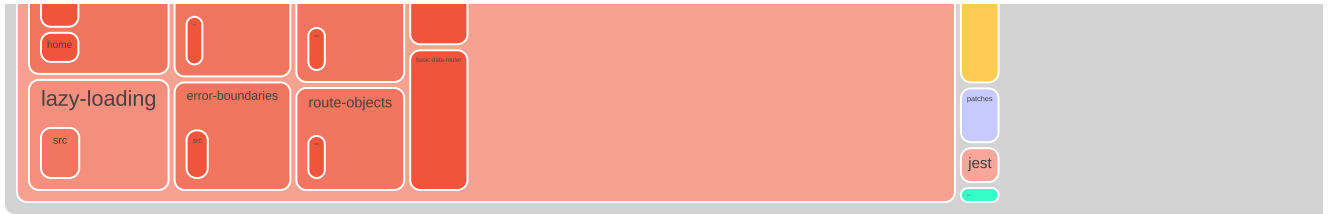
	directoryPath	directoryParentPath	directoryName	fileCount	firs
0	react-router-6.30.0/examples/custom-query-parsing/types	react-router-6.30.0/examples/custom-query-parsing	types	1	react-router-6.30.0/examples/custom-query-parsing/types/jsur
1	react-router-6.30.0/examples/data-router/src	react-router-6.30.0/examples/data-router	src	1	react-router-6.30.0/examples/data-router/src/la
2	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app/routes	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app	routes	1	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/app/routes/_inde
3	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/public	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template	public	1	react-router-6.30.0/integration/helpers/cloudflare-dev-proxy-template/public/favicc
4	react-router-6.30.0/integration/helpers/vite-5-template/app/routes	react-router-6.30.0/integration/helpers/vite-5-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-5-template/app/routes/_inde
5	react-router-6.30.0/integration/helpers/vite-5-template/public	react-router-6.30.0/integration/helpers/vite-5-template	public	1	react-router-6.30.0/integration/helpers/vite-5-template/public/favicc
6	react-router-6.30.0/integration/helpers/vite-6-template/app/routes	react-router-6.30.0/integration/helpers/vite-6-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-6-template/app/routes/_inde
7	react-router-6.30.0/integration/helpers/vite-6-template/public	react-router-6.30.0/integration/helpers/vite-6-template	public	1	react-router-6.30.0/integration/helpers/vite-6-template/public/favicc
8	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app/routes	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/app/routes/_inde
9	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/public	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template	public	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/public/favicc
10	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/workers	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template	workers	1	react-router-6.30.0/integration/helpers/vite-plugin-cloudflare-template/workers/a
11	react-router-6.30.0/integration/helpers/vite-rolldown-template/app/routes	react-router-6.30.0/integration/helpers/vite-rolldown-template/app	routes	1	react-router-6.30.0/integration/helpers/vite-rolldown-template/app/routes/_inde
12	react-router-6.30.0/integration/helpers/vite-rolldown-template/public	react-router-6.30.0/integration/helpers/vite-rolldown-template	public	1	react-router-6.30.0/integration/helpers/vite-rolldown-template/public/favicc
13	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app/routes	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app	routes	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/app/routes/hom
14	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/public	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic	public	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/basic/public/favicc
15	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/blank	react-router-6.30.0/packages/create-react-router/__tests__/fixtures	blank	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/blank/package
16	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/with-ignored-dir	react-router-6.30.0/packages/create-react-router/__tests__/fixtures	with-ignored-dir	1	react-router-6.30.0/packages/create-react-router/__tests__/fixtures/with-ignored-dir/package
17	react-router-6.30.0/packages/react-router-architect/sessions	react-router-6.30.0/packages/react-router-architect	sessions	1	react-router-6.30.0/packages/react-router-architect/sessions/arcTableSessionStora
18	react-router-6.30.0/packages/react-router-cloudflare/sessions	react-router-6.30.0/packages/react-router-cloudflare	sessions	1	react-router-6.30.0/packages/react-router-cloudflare/sessions/workersKVStora
19	react-router-6.30.0/packages/react-router-config	react-router-6.30.0/packages	react-router-config	1	react-router-6.30.0/packages/react-router-config/LIC
20	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app/routes	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app	routes	1	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/app/routes/_inde
21	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/public	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic	public	1	react-router-6.30.0/packages/react-router-dev/__tests__/fixtures/basic/public/favicc
22	react-router-6.30.0/packages/react-router-dev/vite/static	react-router-6.30.0/packages/react-router-dev/vite	static	1	react-router-6.30.0/packages/react-router-dev/vite/static/refresh-uti
23	react-router-6.30.0/packages/react-router-dom/docs	react-router-6.30.0/packages/react-router-dom	docs	1	react-router-6.30.0/packages/react-router-dom/docs/README
24	react-router-6.30.0/packages/react-router-express/__tests__	react-router-6.30.0/packages/react-router-express	__tests__	1	react-router-6.30.0/packages/react-router-express/__tests__/server-ti
25	react-router-6.30.0/packages/react-router-native/docs	react-router-6.30.0/packages/react-router-native	docs	1	react-router-6.30.0/packages/react-router-native/docs/README
26	react-router-6.30.0/packages/react-router-node/sessions	react-router-6.30.0/packages/react-router-node	sessions	1	react-router-6.30.0/packages/react-router-node/sessions/fileStora
27	react-router-6.30.0/packages/react-router/__tests__/__snapshots__	react-router-6.30.0/packages/react-router/__tests__	__snapshots__	1	react-router-6.30.0/packages/react-router/__tests__/__snapshots__/route-matc test.tsx.
28	react-router-6.30.0/packages/react-router/__tests__/dom/components	react-router-6.30.0/packages/react-router/__tests__/dom	components	1	react-router-6.30.0/packages/react-router/__tests__/dom/components/LazyComponer

	directoryPath	directoryParentPath	directoryName	fileCount	firs
29	react-router-6.30.0/packages/react-router/__tests__/dom/polyfills	react-router-6.30.0/packages/react-router/__tests__/dom	polyfills	1	react-router-6.30.0/packages/r outer/__tests__/dom/polyfills/drop-Forml submit

Number of files per directory

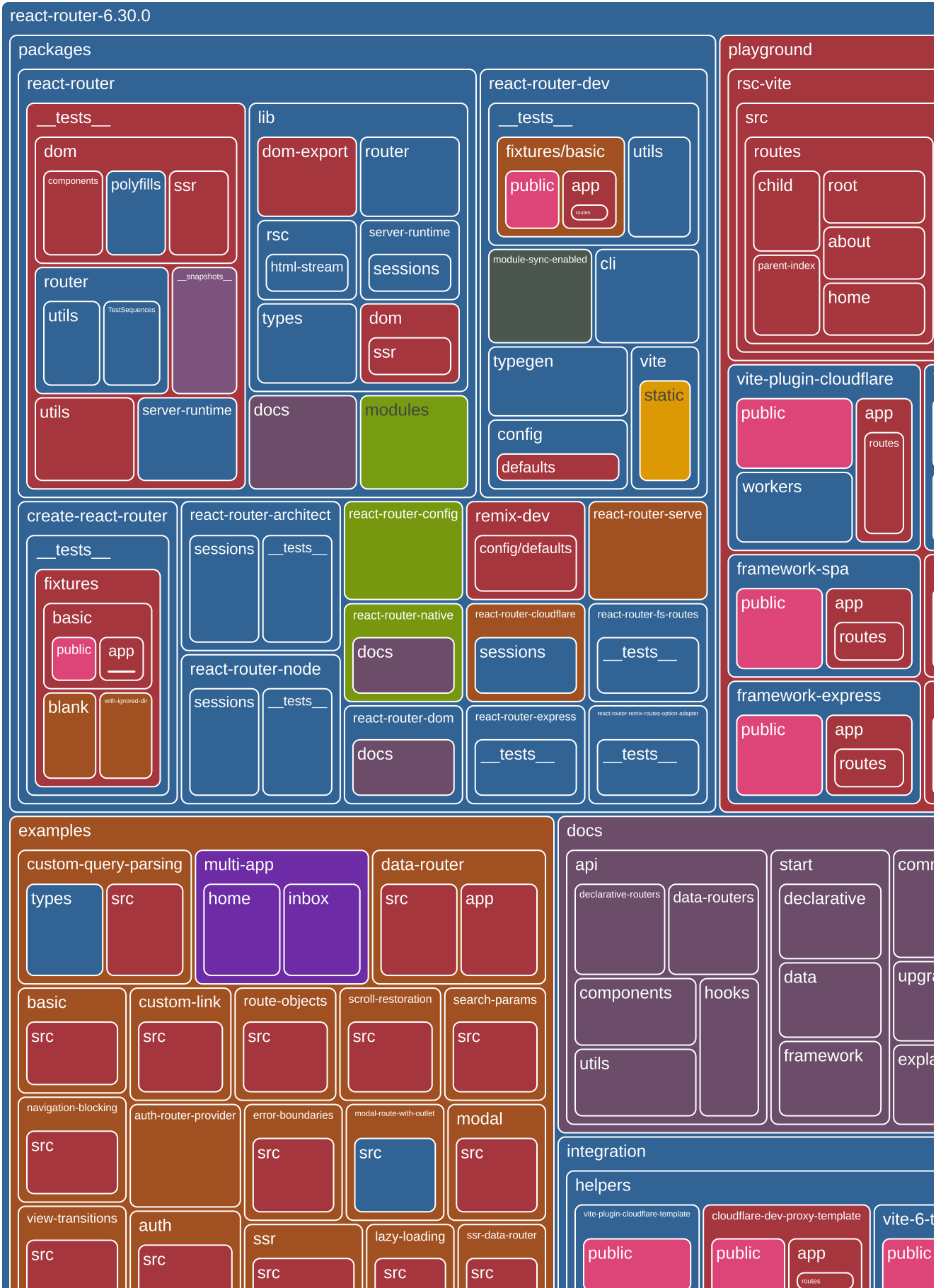
Directories and their file count

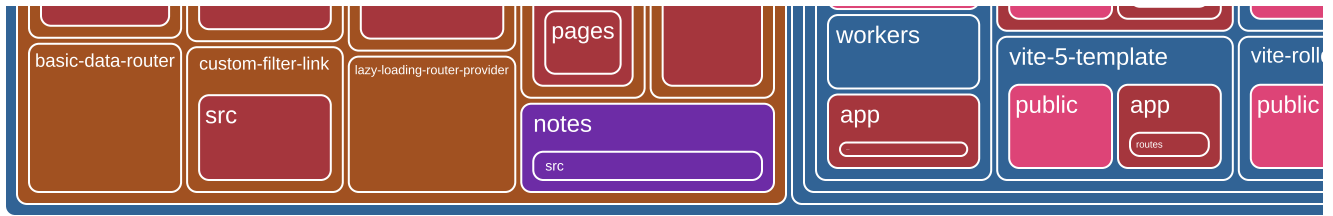




Most frequent file extension per directory

Most frequent file extension per directory

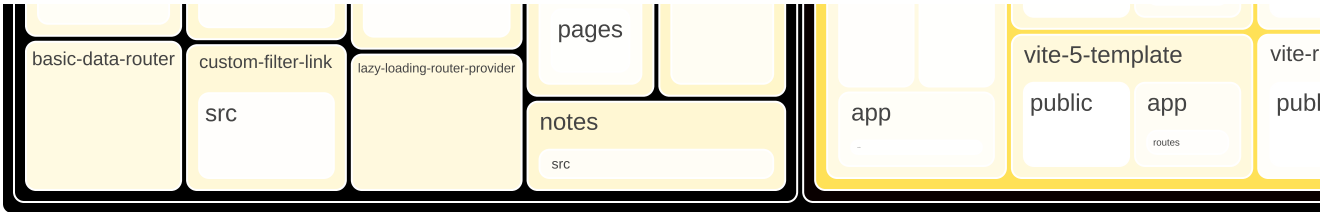




Number of commits per directory

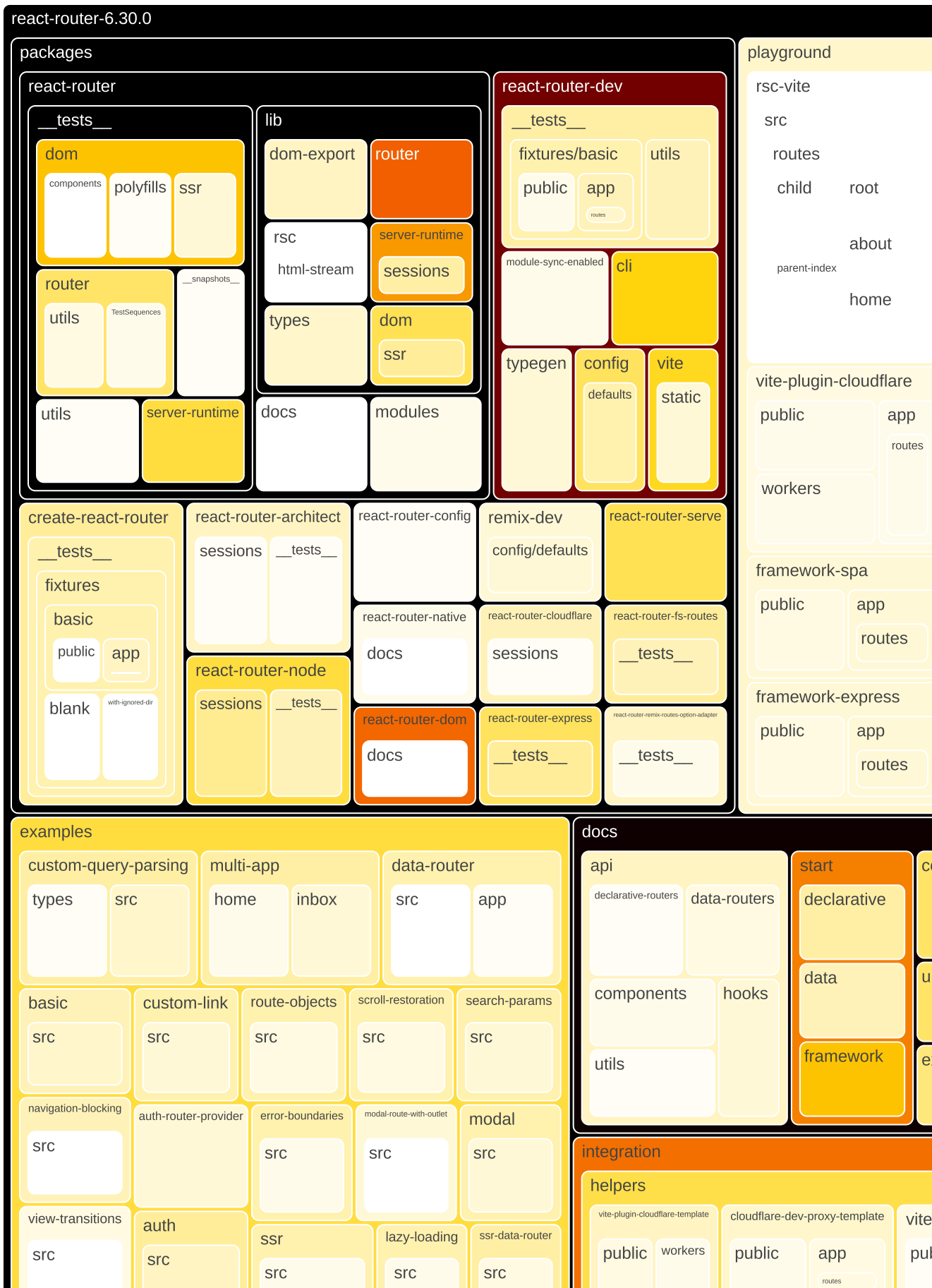
Number of git commits

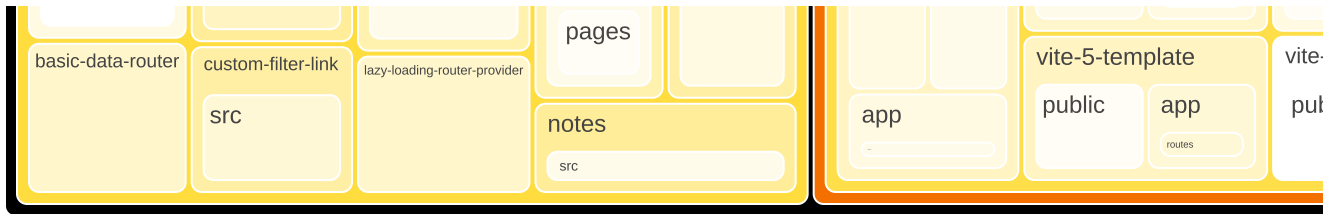




Number of distinct authors per directory

Number of distinct commit authors





Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)

react-router-6.30.0

packages

react-router

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react-router-remix-routes-option-adapter

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__tests__

__tests__

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auth-router-provider

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auth

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error-boundaries

src

ssr

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modal-route-with-outlet

src

lazy-loading

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modal

src

ssr-data-router

src

integration

helpers

vite-plugin-cloudflare-template

public

cloudflare-dev-proxy-template

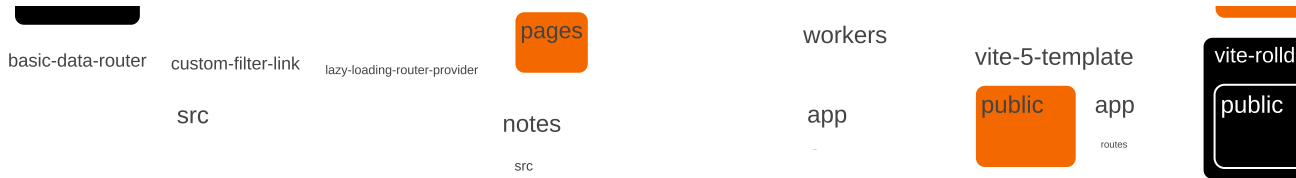
public

app

routes

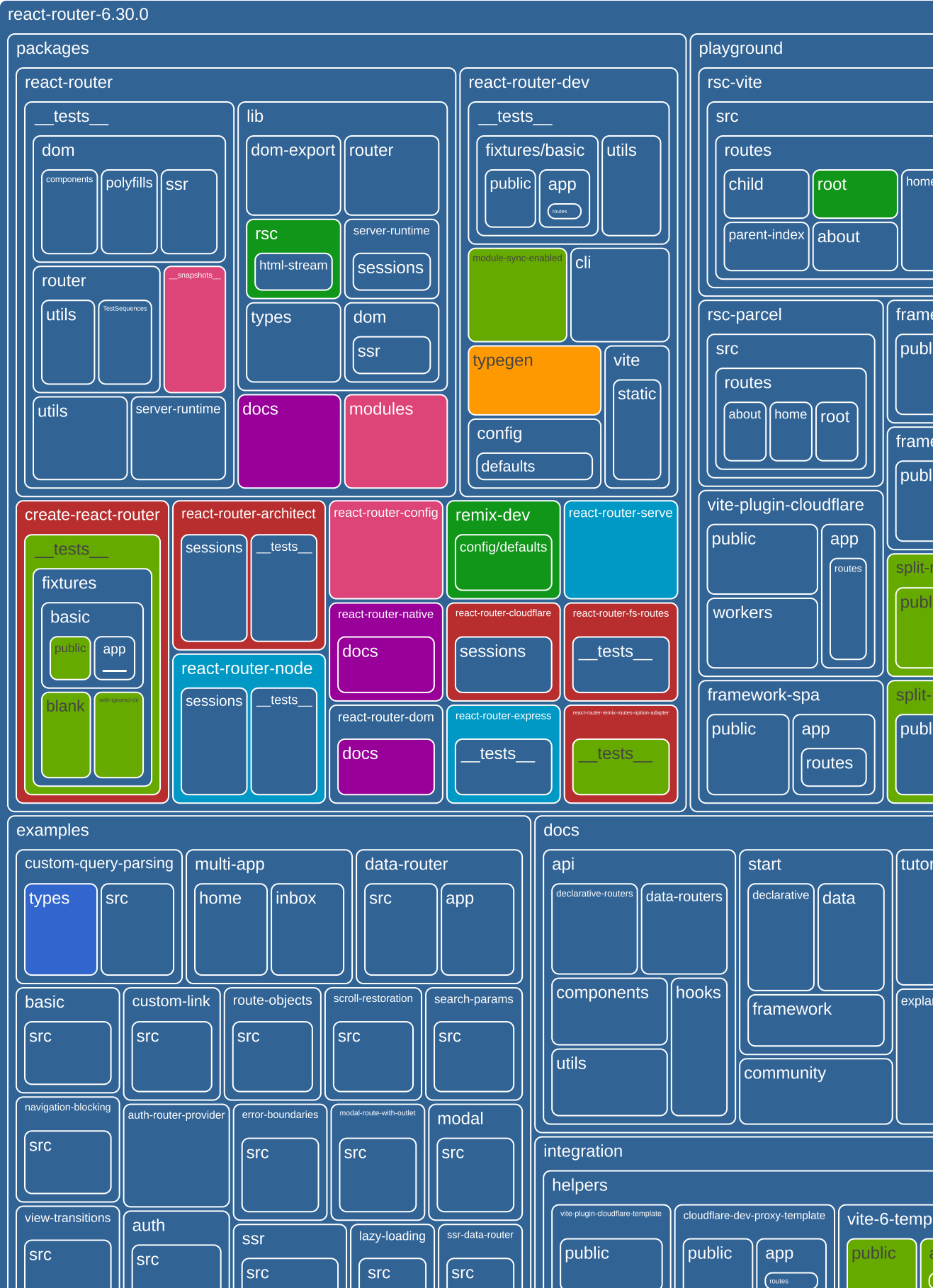
vite-6-te

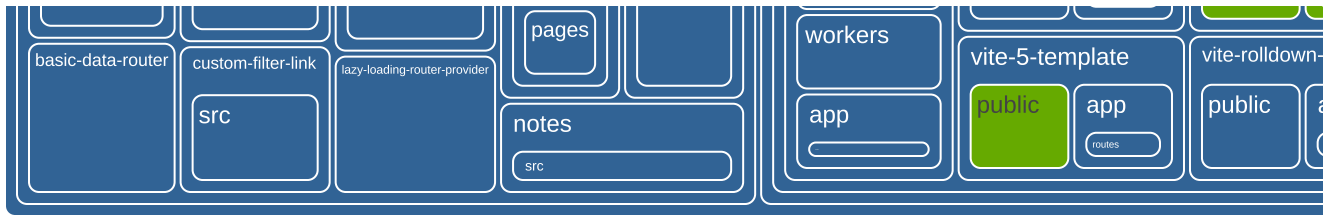
public



Main author per directory

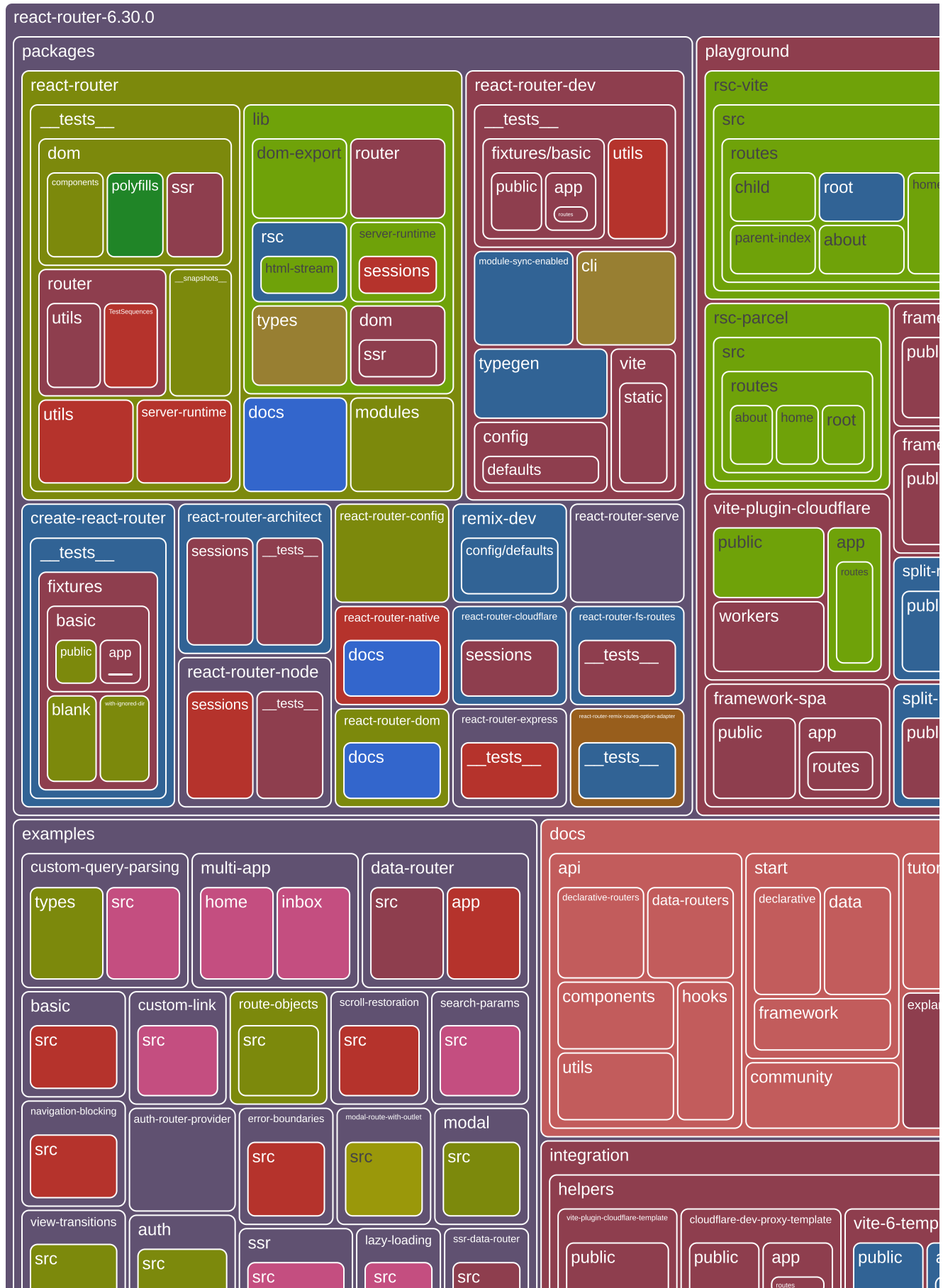
Main authors with highest number of commits

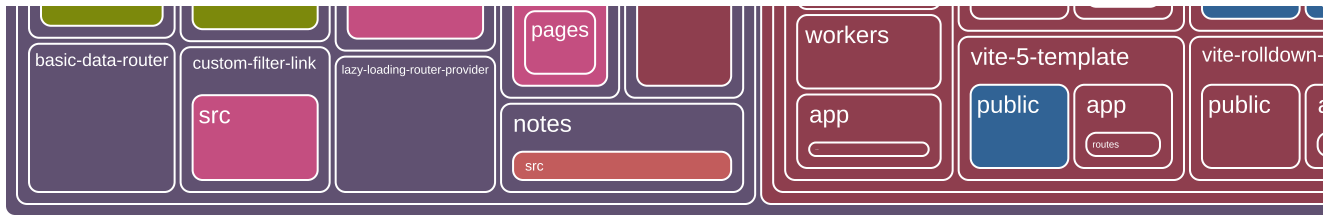




Second author per directory

Second author with the second highest number of commits



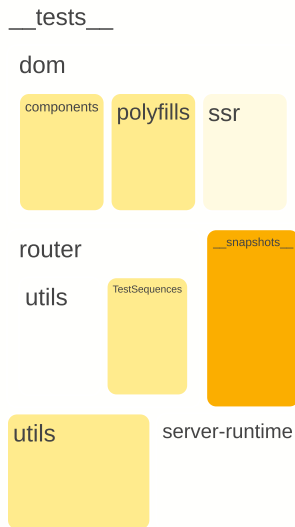


Days since last commit per directory

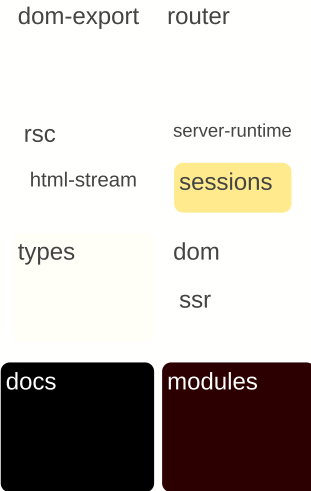
react-router-6.30.0

packages

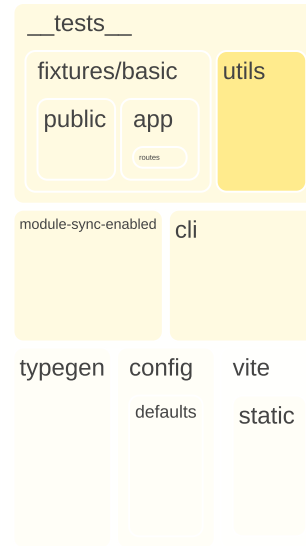
react-router



lib

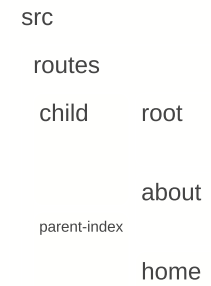


react-router-dev

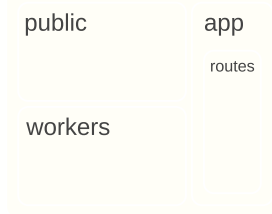


playground

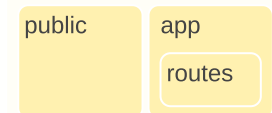
rsc-vite



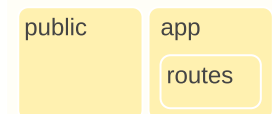
vite-plugin-cloudflare



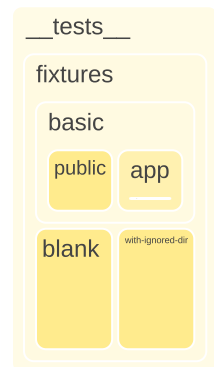
framework-spa



framework-express



create-react-router



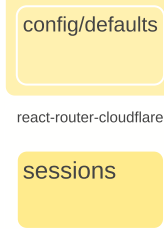
react-router-architect



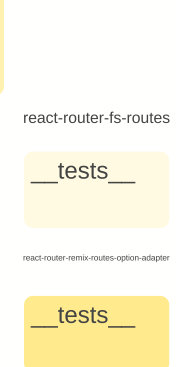
react-router-config



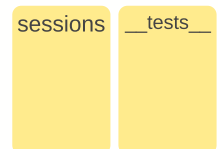
remix-dev



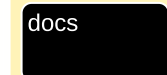
react-router-serve



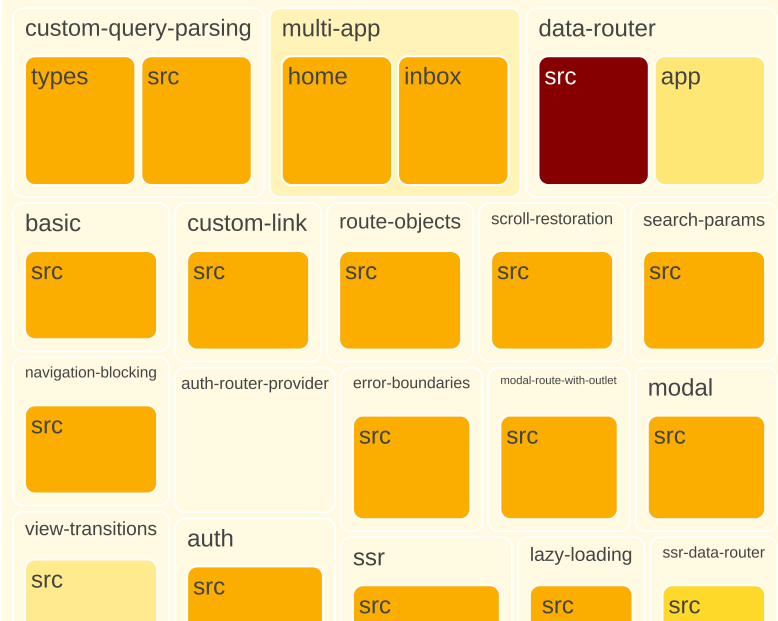
react-router-node



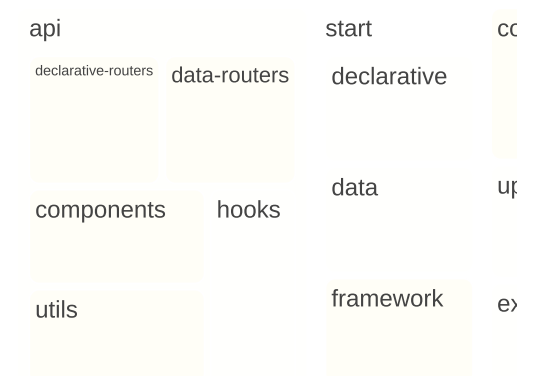
react-router-dom



examples



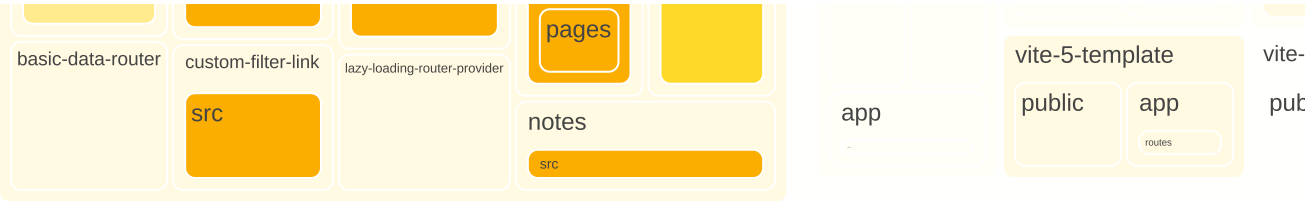
docs



integration

helpers



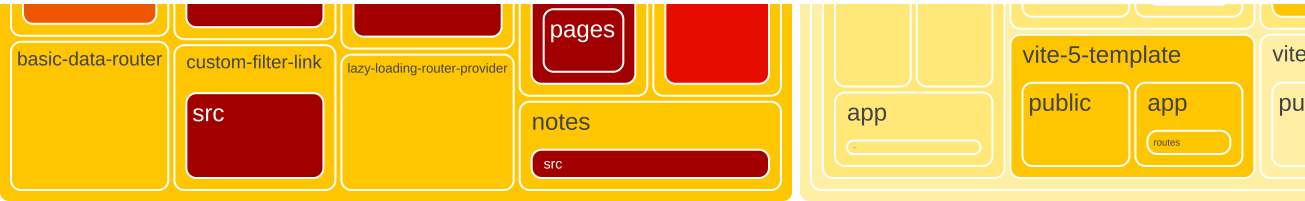


Days since last commit per directory (ranked)

Rank of days since last commit

react-router-6.30.0





Days since last file creation per directory

Days since last file creation

react-router-6.30.0

packages

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react-router-fs-routes

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react-router-remix-routes-option-adapter

__tests__

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declarative

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framework

integration

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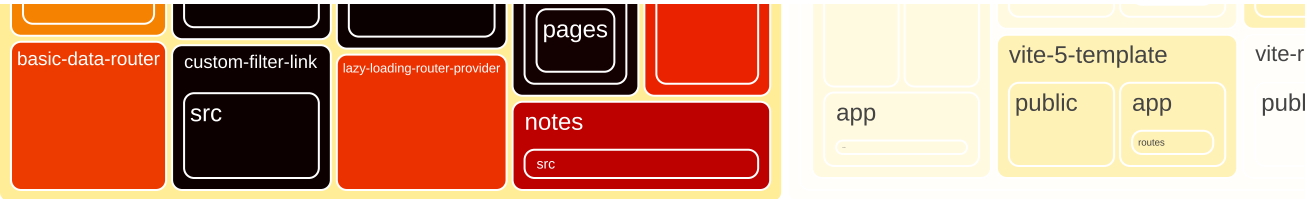
public

app

routes

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Days since last file creation per directory (ranked)

Rank of days since last file creation

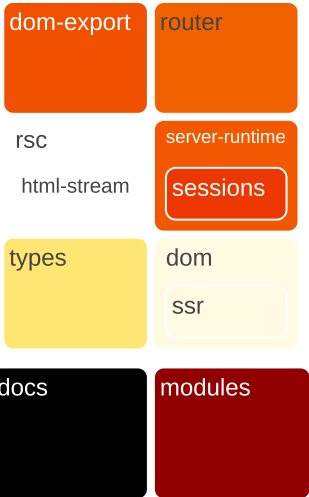
react-router-6.30.0

packages

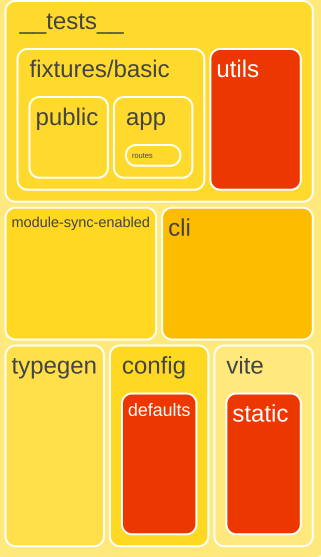
react-router



lib

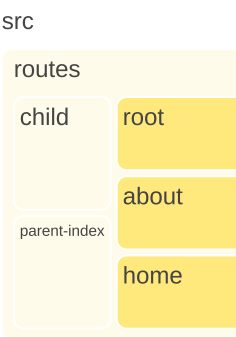


react-router-dev

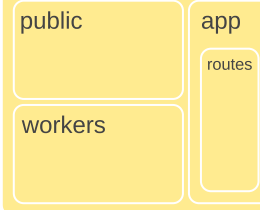


playground

rsc-vite



vite-plugin-cloudflare



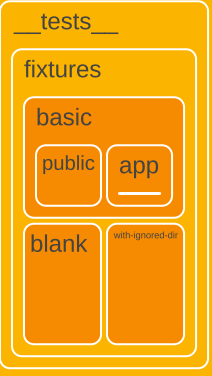
framework-spa



framework-express



create-react-router



react-router-architect



react-router-config



remix-dev



react-router-serve



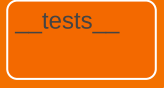
react-router-native



react-router-cloudflare



react-router-fs-routes



react-router-node



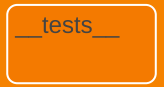
react-router-dom



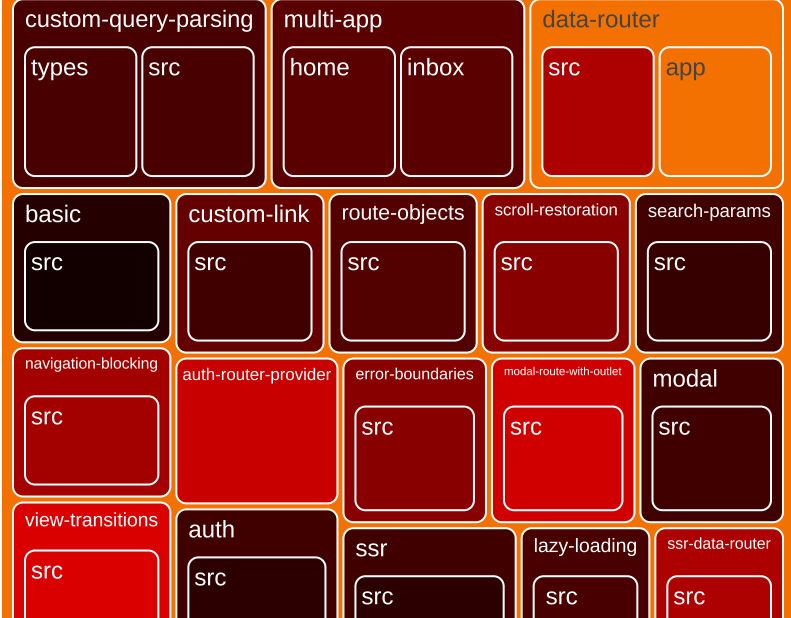
react-router-express



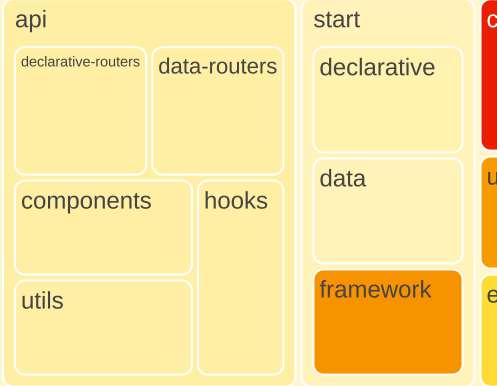
react-router-remix-routes-option-adapter



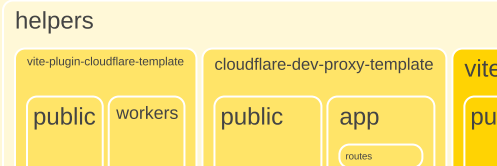
examples

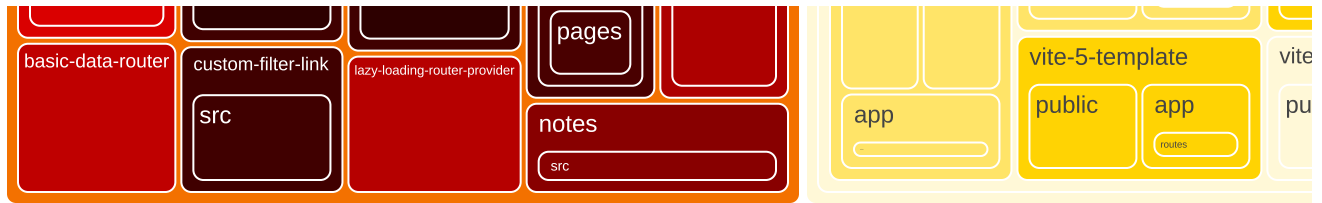


docs



integration

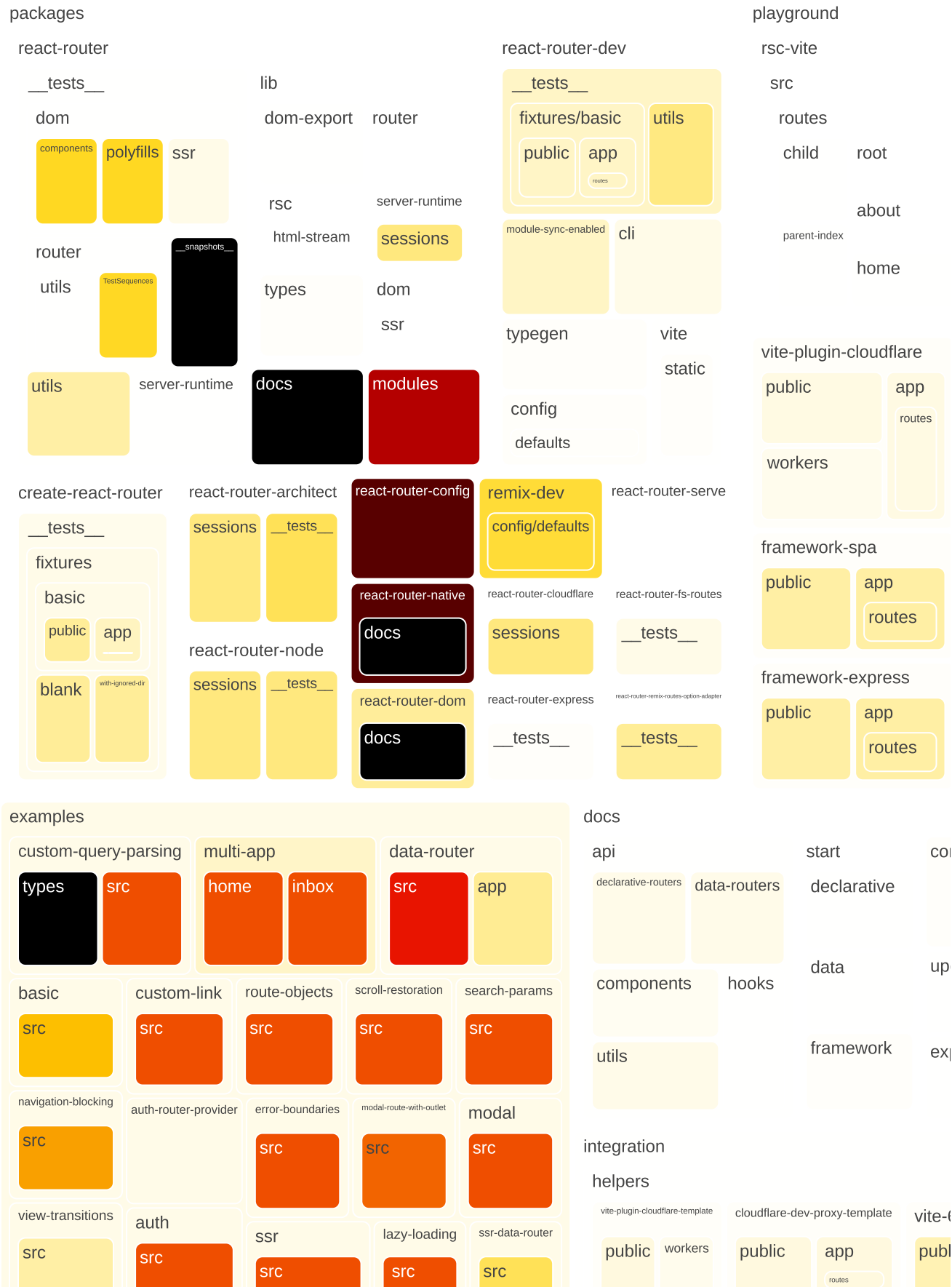


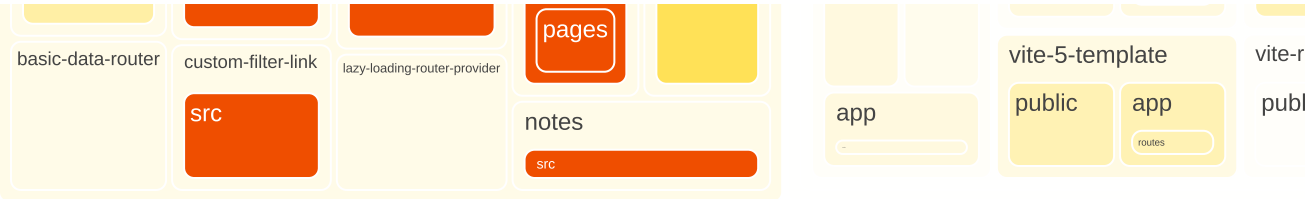


Days since last file modification per directory

Days since last file modification

react-router-6.30.0





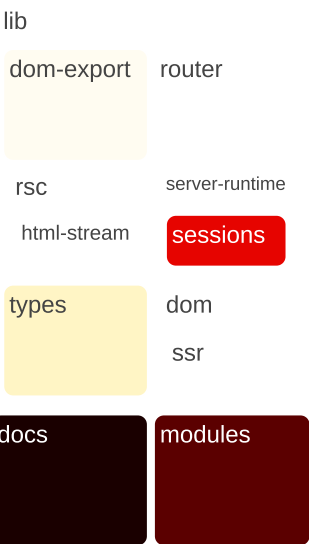
Days since last file modification per directory (ranked)

Rank of days since last file modification

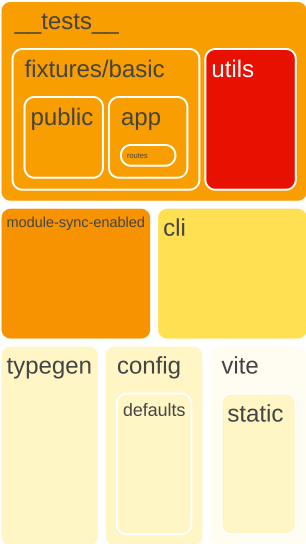
react-router-6.30.0

packages

react-router

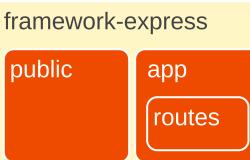
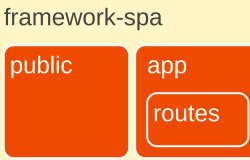
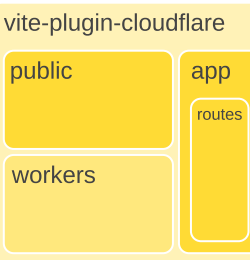
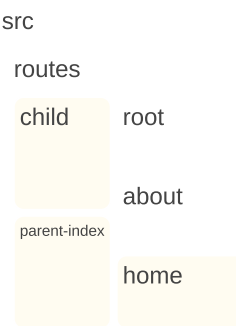


react-router-dev

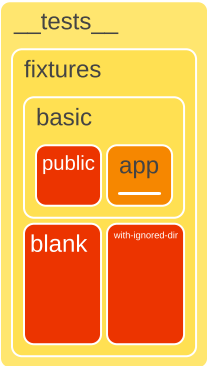


playground

rsc



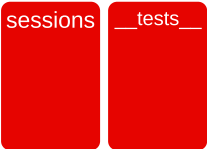
create-react-router



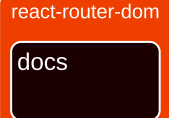
react-router-architect



react-router-node



react-router-config



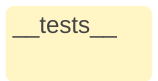
remix-dev



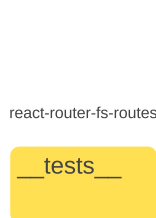
react-router-cloudflare



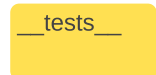
react-router-express



react-router-serve



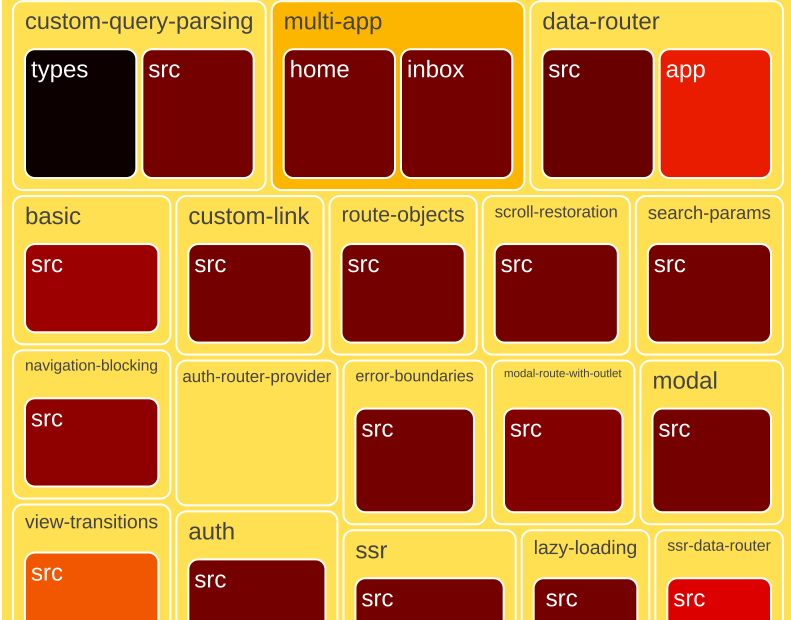
react-router-fs-routes



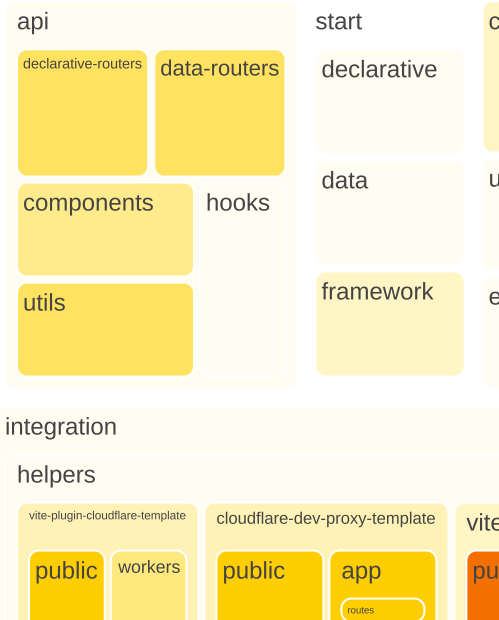
react-router-remix-routes-option-adapter

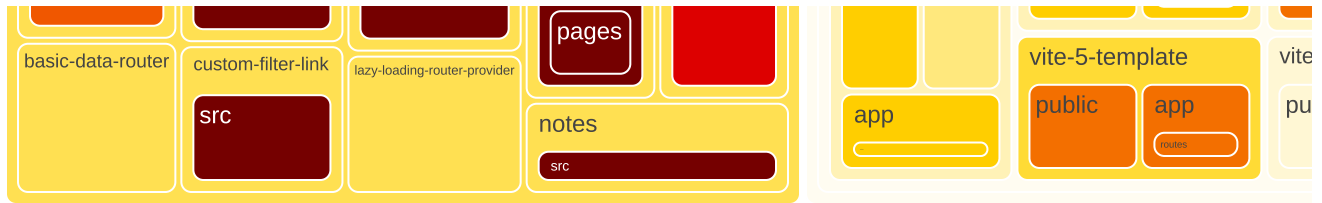


examples



docs

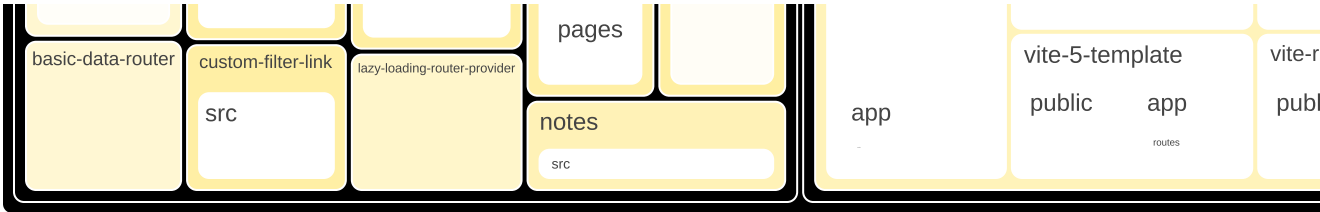




File changed frequently with other files

Pairwise file changes





Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

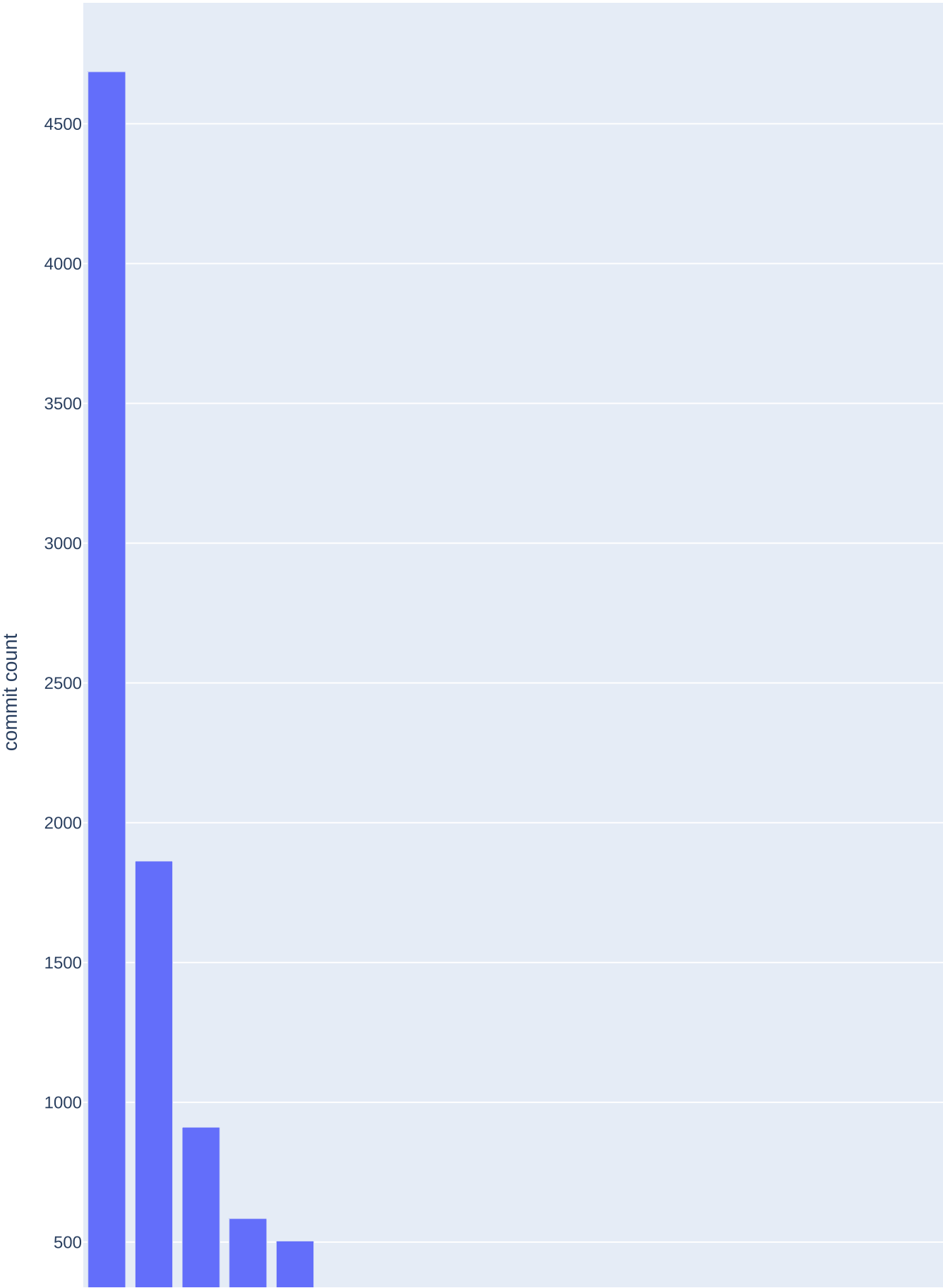
Sum of commits that changed more than 30 files (each) = 482
Max changed files with one commit = 1061

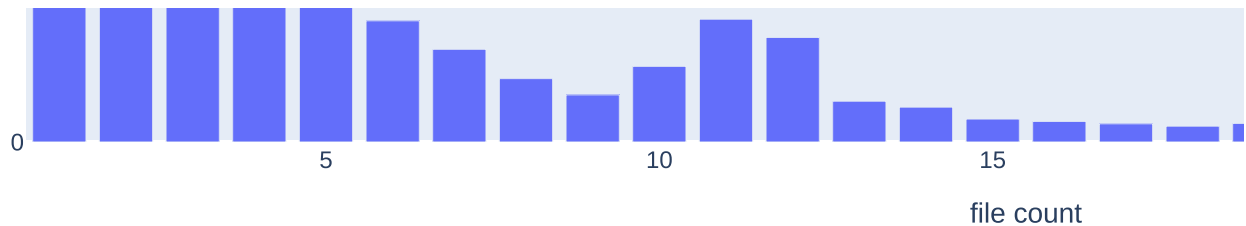
	filesPerCommit	commitCount
count	174.000000	174.000000
mean	152.936782	65.701149
std	193.444744	391.697805
min	1.000000	1.000000
25%	44.250000	1.000000
50%	89.000000	3.000000
75%	163.750000	8.000000
max	1061.000000	4686.000000

	filesPerCommit	commitCount
0	1	4686
1	2	1863
2	3	911
3	4	584
4	5	504
5	6	306
6	7	234
7	8	160
8	9	119
9	10	191
10	11	310
11	12	264
12	13	103
13	14	88
14	15	58
15	16	52
16	17	46
17	18	40
18	19	47
19	20	33
20	21	37
21	22	31
22	23	57
23	24	31
24	25	30
25	26	15
26	27	56
27	28	57
28	29	19
29	30	18

Bar chart with the number of files per commit distribution

Changed files per commit





Pairwise Changed Files vs. Dependency Weight

This section explores the correlation between how often pairs of files are changed together (common commit count) and their dependency weight. Note that these results should be interpreted cautiously, as comparing pairwise changes and dependencies is inherently challenging.

Considerations

- **Historical vs. Current State:** Pairwise changes reflect the entire git history, while dependency weight represents the current state of the codebase.
- **Commit Granularity:** Developers may use different commit strategies, such as squashing changes into a single commit or creating fine-grained commits. Ideally, each commit should represent a single semantic change for accurate analysis.
- **Dependency Representation:** Some file types (e.g., Java files with import statements) clearly define dependencies, while others (e.g., shell scripts, XML, YAML) lack explicit dependency relationships.
- **Repository Characteristics:** Repositories with generated code may have many large commits, while stabilized repositories may only update configuration files for dependency changes.

Data Preview

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: weight)} {position: line: 9, column: 29, offset: 557} for query: "

```
//
List pair of files that were changed together and that have a declared dependency between each other.\n \n MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n //De-duplicating the pairs of files isn't necessary, because the dependency relation is directed.\n //WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n WITH firstCodeFile.fileName AS firstFileName\n      ,secondCodeFile.fileName AS secondFileName\n      ,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n      ,pairwiseChange.commitCount AS commitCount\n      ,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\n RETURN dependencyWeight\n      ,commitCount\n      ,fileDistanceAsFewestChangeDirectoryCommands\n      // ,count(*) AS occurrences\n // ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\n ORDER BY dependencyWeight, commitCount\n \n // MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n // MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n // WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n \n // RETURN firstCodeFile.fileName AS firstFileName\n //      ,secondCodeFile.fileName AS secondFileName\n //      ,dependency.weight AS dependencyWeight\n //      ,pairwiseChange.commitCount AS commitCount\n // ORDER BY dependencyWeight, commitCount\n \n // MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File)\n // WITH count(DISTINCT relation) AS relatedFilesCount\n // ,collect(DISTINCT relation) AS relations\n // UNWIND relations AS relation\n // WITH relatedFilesCount\n //      ,coalesce(relation.commitCount, 0) AS commitCount\n //      ,coalesce(relation.weight, 0) AS dependencyWeight\n //      ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n // RETURN dependencyWeight\n //      ,commitCount\n //      ,fileDistanceAsFewestChangeDirectoryCommands\n // ORDER BY dependencyWeight, commitCount\n"
```

```
Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: fileDistanceAsFewestChangeDirectoryCommands)} {position: line: 11, column: 20, offset: 682} for query: "// List pair of files that were changed together and that have a declared dependency between each other.\n\nMATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\nMATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n//De-duplicating the pairs of files is n't necessary, because the dependency relation is directed.\n//WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\nWITH firstCodeFile.fileName AS firstFileName\n      ,secondCodeFile.fileName AS secondFileName\n      ,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n      ,pairwiseChange.commitCount AS commitCount\n      ,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\nRETURN dependencyWeight\n      ,commitCount\n      ,fileDistanceAsFewestChangeDirectoryCommands\n      // ,count(*) AS occurrences\n      // ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\nORDER BY dependencyWeight, commitCount\n\n// MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n// MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n// WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n// RETURN firstCodeFile.fileName AS firstFileName\n//      ,secondCodeFile.fileName AS secondFileName\n//      ,dependency.weight AS dependencyWeight\n//      ,pairwiseChange.commitCount AS commitCount\n// ORDER BY dependencyWeight, commitCount\n\n// MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File)\n// WITH count(DISTINCT relation) AS relatedFilesCount\n//      ,collect(DISTINCT relation) AS relations\n// UNWIND relations AS relation\n// WITH relatedFilesCount\n//      ,coalesce(relation.commitCount, 0) AS commitCount\n//      ,coalesce(relation.weight, 0) AS dependencyWeight\n//      ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n// RETURN dependencyWeight\n//      ,commitCount\n//      ,fileDistanceAsFewestChangeDirectoryCommands\n// ORDER BY dependencyWeight, commitCount"
```

```
dependencyWeight  commitCount  fileDistanceAsFewestChangeDirectoryCommands
```

Data Statistics

'Pairwise changed git files compared to dependency weights - Overall statistics'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
count	0	0	0
unique	0	0	0
top	NaN	NaN	NaN
freq	NaN	NaN	NaN

'Pairwise changed git files compared to dependency weights - Pearson Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	NaN	NaN	NaN
commitCount	NaN	NaN	NaN
fileDistanceAsFewestChangeDirectoryCommands	NaN	NaN	NaN

'Pairwise changed git files compared to dependency weights - Spearman Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	NaN	NaN	NaN
commitCount	NaN	NaN	NaN
fileDistanceAsFewestChangeDirectoryCommands	NaN	NaN	NaN

Less than 5 samples are not enough to calculate p-values

No data to plot

WordCloud of git authors

	word	frequency
0	Matt Brophy	2131
1	Michael Jackson	1951
2	Ryan Florence	1307
3	Remix Run Bot	531
4	Chance Strickland	511
5	Mark Dalgleish	427
6	Pedro Cattori	412
7	Tim Dorr	392
8	Jimmy Jia	389
9	Jacob Ebey	329

Wordcloud of git authors

